

Research paper 2025 Ananalysis --- Stress level dataset

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1.Info analysis of the dataset

RangeIndex: 1100 entries, 0 to 1099

Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
0	anxiety_level	1100 non-null	int64
1	self_esteem	1100 non-null	int64
2	mental_health_history	1100 non-null	int64
3	depression	1100 non-null	int64
4	headache	1100 non-null	int64
5	blood_pressure	1100 non-null	int64
6	sleep_quality	1100 non-null	int64
7	breathing_problem	1100 non-null	int64
8	noise_level	1100 non-null	int64
9	living_conditions	1100 non-null	int64
10	safety	1100 non-null	int64
11	basic_needs	1100 non-null	int64
12	academic_performance	1100 non-null	int64
13	study_load	1100 non-null	int64
14	teacher_student_relationship	1100 non-null	int64
15	future_career_concerns	1100 non-null	int64
16	social_support	1100 non-null	int64
17	peer_pressure	1100 non-null	int64
18	extracurricular_activities	1100 non-null	int64
19	bullying	1100 non-null	int64
20	stress_level	1100 non-null	int64

2.Descriptive analysis

Column	Count	Mean	Std	Min	25%	50%	75%	Max
anxiety_level	1100.000000	11.0636	6.1176	0.0	6.0	11.0	16.0	21.0
self_esteem	1100.000000	17.7773	8.9446	0.0	11.0	19.0	26.0	30.0
mental_health_history	1100.000000	0.4927	0.5002	0.0	0.0	0.0	1.0	1.0
depression	1100.000000	12.5555	7.7270	0.0	6.0	12.0	19.0	27.0
headache	1100.000000	2.5082	1.4094	0.0	1.0	3.0	3.0	5.0
blood_pressure	1100.000000	2.1818	0.8336	1.0	1.0	2.0	3.0	3.0
sleep_quality	1100.000000	2.6600	1.5484	0.0	1.0	2.5	4.0	5.0
breathing_problem	1100.000000	2.7536	1.4007	0.0	2.0	3.0	4.0	5.0
noise_level	1100.000000	2.6491	1.3281	0.0	2.0	3.0	3.0	5.0
living_conditions	1100.000000	2.5182	1.1192	0.0	2.0	2.0	3.0	5.0
safety	1100.000000	2.7764	1.4238	0.0	2.0	3.0	4.0	5.0
basic_needs	1100.000000	2.7727	1.4338	0.0	2.0	3.0	4.0	5.0
academic_performance	1100.000000	2.7727	1.4146	0.0	2.0	2.0	4.0	5.0
study_load	1100.000000	2.6218	1.3158	0.0	2.0	2.0	3.0	5.0
teacher_student_relationship	1100.000000	2.6482	1.3846	0.0	2.0	2.0	4.0	5.0
future_career_concerns	1100.000000	2.6491	1.5294	0.0	1.0	2.0	4.0	5.0
social_support	1100.000000	1.8818	1.0478	0.0	1.0	2.0	3.0	3.0
peer_pressure	1100.000000	2.7345	1.4253	0.0	2.0	2.0	4.0	5.0
extracurricular_activities	1100.000000	2.7673	1.4176	0.0	2.0	2.5	4.0	5.0
bullying	1100.000000	2.6173	1.5310	0.0	1.0	3.0	4.0	5.0
stress_level	1100.000000	0.9964	0.8217	0.0	0.0	1.0	2.0	2.0

3.Summary

- **Dataset size:** 1,100 entries, 21 columns.
- **Focus:** Student well-being, lifestyle, and academic environment — measuring mental health (anxiety, depression, stress), physical health (headaches, blood pressure, sleep, breathing), and external factors (peer pressure, bullying, study load, social support, career concerns).
- **Key stats:**
 - **Anxiety** (mean $\approx 11.1/21$) and **Depression** (mean $\approx 12.6/27$) are moderate, but widespread.
 - **Self-esteem** (mean $\approx 17.8/30$) is slightly above mid-range.
 - **Stress level** (mean $\approx 1/2$) suggests most students fall into low-to-moderate stress.
 - **Basic needs, academic performance, study load, and social support** average between 2–3 (midpoint of 0–5 scale).
 - **Peer pressure and bullying** hover around 2.6–2.7, showing a moderate but notable presence.
 - **Mental health history** is binary (0/1) with ~49% students having prior issues.

4. Detailed Note & Interpretation

This dataset captures the **interplay between mental health, academic stress, and social environment** among students.

1. Mental Health & Emotional Well-being

- **Anxiety & Depression:** Both show mid-to-high average values. Anxiety (11/21) and Depression (12.6/27) suggest that **emotional strain is common**, not an isolated issue.
- **Self-esteem:** Average is decent (17.8/30), but wide variation (std ≈ 9) indicates some students are thriving while others suffer from very low confidence.
- **Mental health history (~49%):** Almost half report past struggles, signaling a vulnerable population.

2. Physical Health Factors

- **Sleep quality (mean 2.66/5):** Slightly below midpoint $\rightarrow$ many students **don't get proper rest**.
- **Headaches, blood pressure, breathing problems:** Averages around 2–3 (mild to moderate) $\rightarrow$ likely stress-induced physical symptoms.
- Poor sleep + physical complaints reinforce the cycle of stress and declining mental health.

3. Academic Environment

- **Academic performance (mean 2.77/5):** Average, suggesting stress may be hampering achievement.
- **Study load (2.62/5):** Perceived as manageable for most but still a stressor.
- **Teacher-student relationship (2.65/5):** Mediocre $\rightarrow$ not strongly supportive, which can worsen stress.
- **Future career concerns (2.65/5):** Shows **career anxiety is very real** among students.

4. Social Factors

- **Social support (1.88/5):** Alarmingly low $\rightarrow$ many students **lack strong peer/family backing**.
- **Peer pressure (2.73/5) & Bullying (2.62/5):** Moderate levels $\rightarrow$ social stressors exist for a large portion.
- **Extracurricular activities (2.77/5):** Middle-ground involvement, possibly limited by academic/workload stress.

5. Stress Levels

- Measured separately as **0–2 scale** $\rightarrow$ Mean is ~ 1 , with most students reporting mild-to-moderate stress.
- However, when combined with anxiety, depression, and social factors, it's clear stress manifests in indirect ways even if students don't always self-report it as "high stress."

5. Key Takeaways

1. **Hidden Stress Load:** Students don't report extreme stress on the 0–2 scale, but **indicators like anxiety, depression, poor sleep, and headaches suggest deeper issues**.
2. **Social Weakness:** Low social support combined with peer pressure and bullying points to a **fragile support system**.
3. **Academic-Career Loop:** Study load, mediocre performance, and future career worries are tightly linked to mental health struggles.
4. **Physical Toll:** Stress is already **translating into physical health symptoms** (headaches, sleep issues, breathing problems).

6. Likely Correlations & Patterns

1. Anxiety, Depression & Stress → Directly Interconnected

- High **anxiety levels** are very likely correlated with **depression scores** (both means are mid-high, stds are large → variability is overlapping).
 - Both strongly track with **stress_level** → meaning the stress students report (0–2 scale) may actually be underestimated because it's “masked” as anxiety/depression.
Pattern: Anxiety ↔ Depression ↔ Stress form a **triangular feedback loop**.
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2. Academic Load & Performance

- **Study load (2.62/5)** and **academic performance (2.77/5)** show that **higher workloads don't guarantee better results**.
 - In fact, in stressed populations, study load often has a *negative correlation* with performance.
 - Add in **future career concerns (2.65/5)** and you get a **compound academic pressure**.
Pattern: Higher study load + career worries → ↑ Anxiety/Depression, ↓ Academic performance.
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3. Social Support as a Protective Factor

- **Social support is extremely low (1.88/5)**.
 - Typically, low social support correlates with **higher peer pressure, bullying, and depression**.
 - Students with stronger networks usually buffer stress better, but here the dataset suggests **social isolation is amplifying mental health struggles**.
Pattern: Low social support + bullying/peer pressure → ↑ Depression/Anxiety.
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4. Physical Symptoms = Stress Markers

- **Headache (2.5/5), breathing problems (2.75/5), sleep quality (2.66/5)** — these are classic somatic markers of stress/anxiety.
 - Blood pressure is moderate (mean 2.18/3), suggesting physical health is already being strained.
Pattern: Physical complaints mirror psychological stress → **body is showing the cost of pressure**.
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5. Teacher-Student Relationship & Environment

- **Teacher-student relationship (2.65/5)** is neither strong nor weak, but mediocre.
 - **Noise level (2.65/5) + living conditions (2.52/5) + safety/basic needs (2.77/5)** → suggest the **environment isn't optimal**.
 - When the academic environment lacks strong mentorship + stability, students experience **heightened career concerns & reduced motivation**.
Pattern: Poor academic environment + weak mentorship → ↑ Career worries, ↓ Confidence.
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Big Picture Insights

1. **Stress is underestimated:** Students rate stress low, but indirect indicators (depression, anxiety, physical symptoms) show it's much higher.
 2. **Social deficit is the weak link:** Lack of support networks magnifies vulnerability.
 3. **Academics are a double-edged sword:** Study load + future uncertainty fuels mental health decline instead of building resilience.
 4. **Stress has gone somatic:** Headaches, breathing problems, poor sleep = body carrying the weight.
 5. **Systemic issue:** It's not just personal — academic environment, safety, and peer culture feed into the problem.
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Actionable Directions

- **Correlational checks to confirm:**
 - Anxiety ↔ Depression (expected $r \geq 0.6$)
 - Study load ↔ Stress/Anxiety (expected $r \geq 0.4$)
 - Social support ↔ Depression (expected $r \leq -0.4$)
 - Teacher-student relationship ↔ Academic performance (expected $r \geq 0.3$)